





ENTROPY



► When  $\eta(\mathrm{d}x) = \mathrm{d}x$ , we have the relative entropy reduces to the **Shannon entropy**:

$$S(\pi) = -k_B \int_{\mathbb{X}} \pi(x) \log \pi(x) \mathrm{d}x$$

► When  $X$  is discrete, and  $v \in \mathbb{R}$

- Suppose  $\pi$  is the uniform distribution over  $X_u = \{x \in X : U(x) = u\}$

$$\pi(\mathrm{d}x) = \delta_{X_u}(\mathrm{d}x)$$

- We have the entropy of the system measures the number of states with energy  $u$

$$S(\pi_u) = -k_B \log |X_u|$$

Entropy is a measure of disorder in a system with a given energy





$S(\pi) \leq 0$  and  $S(\pi) = 0$  if and only if  $\pi = \eta$

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